Sorting, and judging algorithms

Programming · Level 3 · Lesson 4



In this lesson you will

- Explain bubble, insertion and merge sort.
- Judge algorithms by correctness first, then work.
- Test with edge cases.

Key ideas

- Bubble and insertion: up to about n × n ÷ 2.
- Merge: about n × log n.
- Correct first: empty, one, sorted, reversed, repeats.

Activity: Human sorting

1. Warm-up (3 min): Eight classmates hold number cards in a row.
2. Bubble (7 min): Swap neighbours, pass after pass. Count.
3. Insertion (7 min): Start again; slide each into place. Count.
4. Compare (3 min): Which needed fewer comparisons?

Write the starting order down so both methods start from the same row: a fair race.

Let’s discuss

1. Why count steps instead of seconds?
2. When would insertion sort beat merge sort?
3. Which edge case would you test first, and why?

Exit ticket

- Name three edge cases for a sorting function.